ALICE L WALTON SCHOOL OF MEDICINE

Sideroblastic Anemia

Last updated November 19, 2024 · 14 min read · ScholarRx

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Adapted from: Sideroblastic Anemia

Learning Objectives (4) · Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of the mentioned anemia.
2. Describe the pathophysiology of the mentioned anemia.
3. Interpret clinical findings to properly diagnose the mentioned anemia.
4. Outline the management of the mentioned anemia.

CASE CONNECTION

MH, a 56-year-old male, is a new patient at the gastroenterology clinic. He has alcoholic cirrhosis and was recently admitted for management of ascites. The lab results showed low hemoglobin, low mean corpuscular volume, and high ferritin. The report of the blood smear says: “Rare Pappenheimer bodies seen.”

What other diagnostic test will affirm the preliminary diagnosis? How should this condition be explained to MH? Consider your answers as you read, and we’ll revisit MH at the end of the brick.

GO TO CONCLUSION ↓

What Is Sideroblastic Anemia?

Iron plays a critical role in hemoglobin function and oxygen delivery to the tissues. Even a small decrease in iron levels causes clinically significant impairment in blood oxygen content. This is most often seen in iron deficiency anemia (low iron levels) or anemia of chronic disease (inability to utilize available iron). Both of these anemias cause **ineffective erythropoiesis** (i.e., limited formation of new red blood cells [RBCs] by the bone marrow).

Sideroblastic anemia is a more unusual kind of anemia that causes ineffective erythropoiesis. In this condition, **impaired RBC utilization of iron** prevents normal heme formation, thus limiting formation of normal hemoglobin and normal RBCs. In sideroblastic anemia the unchelated iron accumulates in mitochondria, which are arranged in a ring-like configuration around the nuclei of bone marrow erythroid progenitor cells (erythroblasts), giving these cells the name **ringed sideroblasts.** The name comes from the Greek word sideros, which means

Sideroblastic anemia is not a single disease but instead a pattern of RBC injury that can be caused by a variety of hereditary and acquired diseases.

Why are ringed sideroblasts present in the bone marrow of patients with sideroblastic anemia?

Epidemiology

Hereditary sideroblastic anemia is uncommon and is usually diagnosed in infants and young children. Among infants with anemia, it is worth looking for: in the United States, the prevalence of ringed sideroblasts in the bone marrow is approximately 8% in infants with anemia.

The more common nonhereditary causes seen in adults are due to a variety of conditions. The exact prevalence is unknown since diagnostic bone marrow biopsy is not often done in most of these patients.

How Do Patients with Sideroblastic Anemia Present (LO #1)?

Like most anemias, the symptoms of sideroblastic anemia depend on the severity of anemia, ranging from asymptomatic to dizziness, syncope, and palpitations. Patients may also show other symptoms of the underlying cause: infections, fatigue, bleeding, or bruising in myeloproliferative or myelodysplastic syndromes or episodes of alcohol withdrawal in heavy alcohol use.

Infants and children with the rare hereditary forms of sideroblastic anemia may show a variety of symptoms, such as failure to thrive, developmental delay, immunodeficiency, fevers, myopathy, and lactic acidosis. In these cases, the anemia is part of a larger metabolic disorder.

Mnemonic

The important signs of lead poisoning can be remembered with the following mnemonic:

CLINICAL CORRELATION

A classic presentation mentioned in clinical vignettes of children with lead poisoning is anemic and neurologic symptoms appearing in a child who lives in an old house with chipped paint, as older housing units built prior to 1960 used lead-based paint that gets ingested by young children.

Though lead-based paint has decreased in use, areas of lead exposure remain, including industrial emissions, lead paint in older housing, lead-contaminated soil, water service lines, and consumer products.

How do patients with sideroblastic anemia present?

What Is the Pathophysiology of Sideroblastic Anemia (LO #2)?

Sideroblastic anemia results from **inhibition of the protoporphyrin synthesis** pathway. To understand the various defects in protoporphyrin ring synthesis, we’ll first examine the heme synthesis pathway (Figure 1).

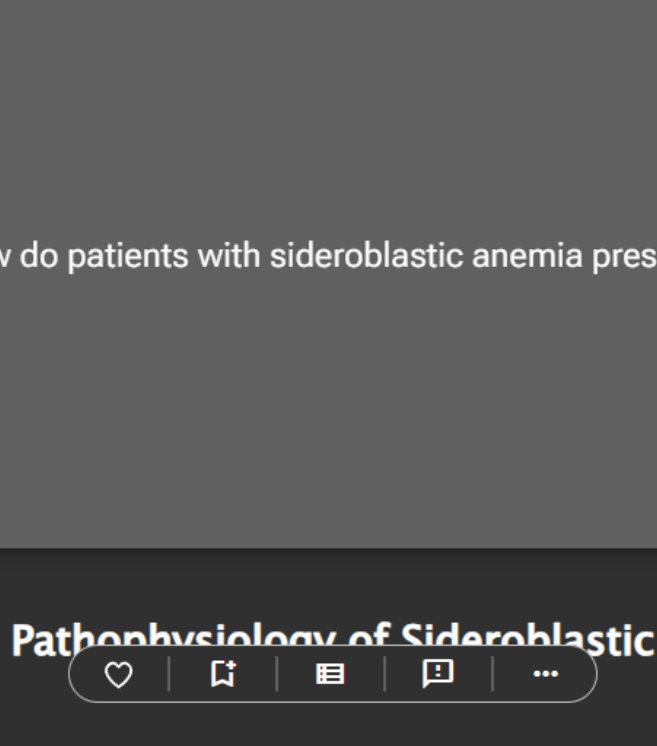

Figure 1

This pathway occurs within erythroblasts in the bone marrow and forms protoporphyrin IX. Macrophages transfer iron to erythroblasts, where it enters the mitochondria and combines with protoporphyrin IX to form heme. Recall that peripherally circulating mature erythrocytes cannot synthesize heme because they do not have mitochondria.

In this eight-step pathway in Figure 1, the first and last steps are catalyzed by the enzymes δ-aminolevulinic acid synthase (ALAS) and ferrochelatase, respectively. These are the enzymes most relevant to this discussion. Note that ALAS requires vitamin B_6 as a coenzyme to function properly.

When protoporphyrin IX is deficient, unchelated iron accumulates within the mitochondria in the erythroblasts, causing a ring to form around the nucleus of erythroid progenitor cells. These **ringed sideroblasts** are the hallmark of sideroblastic anemia and can be seen in the bone marrow.

Note that because of the ineffective iron incorporation into heme, the iron stores in the bone marrow are increased. With so much excess iron, the production of free radicals increases, and RBCs undergo hemolysis within the bone marrow, leading to an increase in total iron and potential iron overload. So, while largely an anemia of ineffective erythropoiesis, sideroblastic anemia sometimes also has a component of hemolytic anemia as well. However, these patients don’t typically have the findings associated with hemolytic anemia like jaundice or increased serum unconjugated bilirubin.

What is the mechanism of iron overload in sideroblastic anemia?

Etiology

Causes of sideroblastic anemia can be categorized into acquired and hereditary causes.

Hereditary Causes. Genetic (congenital) causes of sideroblastic anemia are rare. They involve deficiencies or decreased function of the enzymes that synthesize heme. The most common is a gene mutation that decreases ALAS activity, which is usually due to an X-linked ALAS mutation (see Figure 1). This disorder is therefore most common in males. Because vitamin B_6 is a cofactor for this enzyme, anemia may improve with vitamin B_6 supplementation.

Acquired Causes. Acquired causes are more common; they are due to many different disorders most common in adults. Some of these disorders include:

- Hematologic neoplasms: these can disrupt RBC and other blood cell formation in the bone marrow. Examples include **myelodysplastic syndromes** and **myeloproliferative neoplasms** such as polycythemia vera or chronic myeloid leukemia. Approximately 30-35% of patients with myelodysplastic syndromes have ringed sideroblasts on bone marrow samples.
- Vitamin deficiencies: **vitamin B_6 (pyridoxine) deficiency** causes sideroblastic anemia because the vitamin is needed for ALAS action. It is seen most often erculosis drug isoniazid, and in profound malnutrition. Copper deficiency is an unusual cause and is most often due to excess zinc taken as a supplement (zinc decreases gut copper absorption).
- Toxins and drugs: alcohol, isoniazid (which reduces vitamin B_6 levels), and the antibiotics linezolid and chloramphenicol can inhibit heme synthesis. In addition, **lead poisoning** can cause sideroblastic anemia as lead inhibits the ferrochetalase and ALA dehydratase enzymes.

Why does vitamin B_6 deficiency lead to sideroblastic anemia?

Pathology

The peripheral blood smear and bone marrow samples both can show the abnormal cells that characterize this form of anemia.

Peripheral Blood Sm deposits in the circulating RBCs. The look like clusters of blue granules usually in the periphery of the cell. The arrows in Figure 2 show Pappenheimer bodies on a peripheral blood smear stained with Prussian blue.

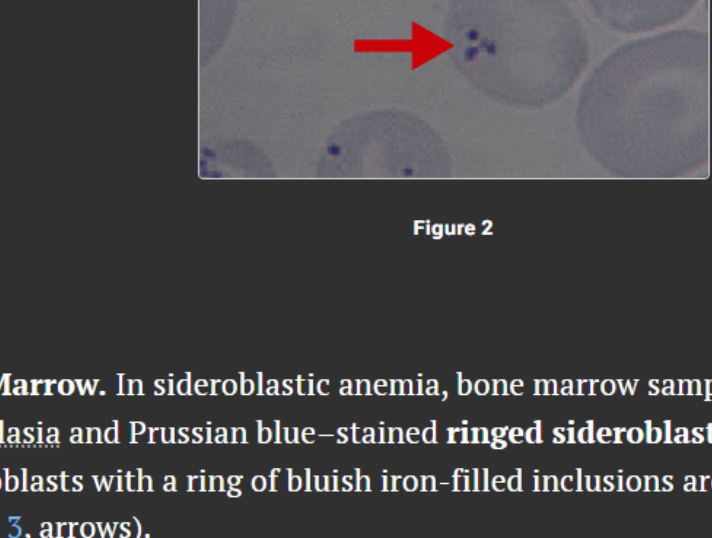

Figure 2

Bone Marrow. In sideroblastic anemia, bone marrow samples display erythroid hyperplasia and Prussian blue–stained **ringed sideroblasts,** seen as erythroblasts with a ring of bluish iron-filled inclusions around their nuclei (Figure 3, arrows).

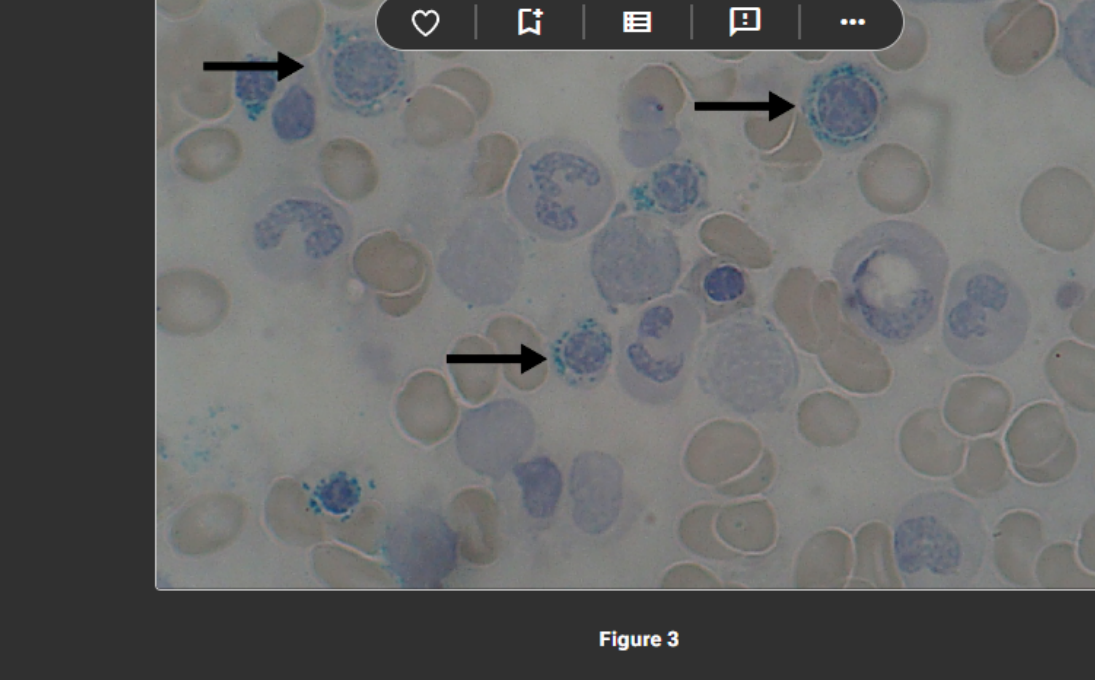

Figure 3

Because ringed sideroblasts are nucleated erythroid progenitor cells, they are present only in the bone marrow and are not seen on peripheral blood smear.

CLINICAL CORRELATION

Because the ineffective erythropoiesis leads to the buildup of excess unused iron, **iron overload** can develop in some of these patients. The findings are similar to those of congenital hemochromatosis and can include cardiac, hepatic, endocrine, and skin disease. Iron overload is most common in the rare hereditary forms of sideroblastic anemia and is related to disease duration. It can be worsened by frequent transfusions given for the anemia.

How Do We Diagnose Sideroblastic Anemia (LO #3)?

Sideroblastic anemia can be considered in any patient with signs and symptoms of anemia (e.g., pallor, weakness, fatigue), especially in the setting of one of the causes listed above like heavy alcohol use or vitamin deficiencies. Lab testing is required for diagnosis.

In patients with a concurrent myeloproliferative or myelodysplastic disorder, also look for signs like splenomegaly and findings of reduced platelets or decreased platelet function (e.g., bleeding, petechiae).

Laboratory Tests

The complete blood count (CBC) in patients with sideroblastic anemia usually shows **microcytic anemia** (i.e., decreased hemoglobin and hematocrit with low mean corpuscular volume [MCV]). However, the MCV may sometimes be normal or high.

Some cases of sideroblastic anemia show dimorphic RBCs (i.e., both normocytic and microcytic cells) on the CBC. The presence of dimorphic RBCs is not pathognomonic for sideroblastic anemia, as it can also be seen in patients with two different types of anemias simultaneously. On the blood smear, look for RBCs with basophilic stippling and Pappenheimer bodies (see Figure 2).

Genetic testing is done for infants and children with suspected hereditary causes to identify the specific gene mutation and guide prognosis and therapy.

Bone Marrow Biops

Confirmation of the diagnosis requires the presence of ringed sideroblasts on microscopic examination of bone marrow aspirate or biopsy sample. Other bone marrow findings include compensatory hyperplasia of erythroid progenitor cells (normoblasts), as they try (futilely) to make up for the ineffective erythropoiesis.

What are typical abnormal findings seen on the blood smear and bone marrow biopsy in sideroblastic anemia?

How Do We Manage Sideroblastic Anemia (LO #4)?

The treatment for non-hereditary (adult) sideroblastic anemia is usually targeted to the underlying cause (e.g., avoidance of alcohol or zinc, replacement of copper or vitamin B_6, or chelation of iron in hereditary hemochromatosis). Myelodysplastic syndromes or myeloproliferative neoplasms should be (e.g., hydroxyurea or other chemotherapy drugs) and or bone marrow transplantation.

Many cases of hereditary sideroblastic anemia respond well to vitamin B_6 supplementation, likely because it activates the first (rate-limiting) step of the protoporphyrin synthesis pathway. Occasionally, a cure can be achieved using allogenic stem cell transplantation. Patients with iron overload should receive chelation (e.g., desfirasirox) therapy to remove excess iron.

In addition, patients should receive supportive anemia treatment as needed, including transfusions.

What is the best treatment of adult (non-hereditary) sideroblastic anemia?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MH, what other diagnostic test can affirm the preliminary diagnosis? How do y

You explain to MH that alcohol consumption damaged not only his liver but also his bone marrow. You explain that he has sideroblastic anemia caused by alcohol-induced disruption in the synthesis of heme, the iron-containing molecule in blood. MH does not need a transfusion at this point, and the focus will be on treatment of his liver disease. Total cessation of alcohol use is recommended, and he is given a vitamin B_6 supplement, as this deficiency can cause sideroblastic anemia. He is provided a list of resources to help him. MH is placed on the liver transplant list and continues to follow up in the gastroenterology (GI) clinic.

Summary

- Sideroblastic anemia is caused by impaired utilization of iron, preventing formation of heme, a critical component of hemoglobin.
- The unchelated iron accumulates in mitochondria, which are arranged in a ring-like configuration around the nucleus of bone marrow erythroid progenitor cells.
- Patients are often asymptomatic, but, if the anemia is severe, they can show common anemia symptoms such as easy fatigability, palpitations, syncope, and exertional dyspnea.
- Infants and children with the unusual hereditary forms may show a variety of symptoms, such as failure to thrive, developmental delay, immunodeficiency, fevers, myopathy, and lactic acidosis.
- Sideroblastic anemia is due to enzyme deficiencies in the protoporphyrin IX synthesis pathway in red blood cell (RBC) precursors, leading to poor incorporation of iron into heme and ineffective erythropoiesis.
- Sideroblastic ane d disorders like heavy alcohol use, or myelodysplastic/myeloproliferative syndromes. An important acquired cause that can be seen in children is lead poisoning, due to inhibition of the ferrochetalase and ALA dehydratase enzymes.
- Hereditary sideroblastic anemia is rare, diagnosed in children, and is caused by enzyme deficiencies in the heme synthesis pathways.
- Patients have signs of anemia such as pallor, tachycardia, and orthostatic hypotension as well as features of underlying cause.
- The blood smear may show Pappenheimer bodies, basophilic stippling, or dimorphic RBCs, while bone marrow biopsy will show ringed sideroblasts.
- Acquired sideroblastic anemia involves treating the underlying disorder or vitamin deficiency.
- Treating hereditary sideroblastic anemia often includes supplementation with vitamin B_6.

Review Questions

1. Which of the following peripheral blood findings is most common in sideroblastic anemia?

A. ○ Ringed sideroblasts
B. ○ Heinz bodies
C. ● Pappenheimer bodies
D. ○ Schistocytes
E. ○ Sickle cel

Hide Explanation

The correct answer is Pappenheimer bodies (C). The diagnosis is confirmed with bone marrow aspirate revealing ringed sideroblasts on Prussian blue stain, but ringed sideroblasts (A) are not present in the peripheral blood. Pappenheimer bodies are the most common peripheral blood smear findings in all causes of sideroblastic anemia and reflect abnormal iron incorporation into the cells. Heinz bodies (B) are seen in G6PD deficiency. Schistocytes (D) are fragmented red blood cells; these are not present in sideroblastic anemia. Sickle cells (E) are present in sickle cell anemia.

2. A 45-year-old male presents to the clinic with fatigue over the past few weeks. His medical history is significant for a recent exposure to tuberculosis for which he was placed on isoniazid. Complete blood count (CBC) reveals decreased hemoglobin and decreased mean corpuscular volume (MCV). Iron studies reveal increased ferritin and decreased total iron-binding capacity. Peripheral blood smear shows red blood cells containing blue inclusions. Which of the following treatments would be most effective in this patient?

A. ○ Deferoxamine
B. ○ Phlebotomy
C. ● Vitamin B_6
D. ○ Vitamin E
E. ○ Folic acid

Hide Explanation

The correct answer is vitamin B_6 (C). This patient presents with fatigue and has microcytic anemia on complete blood count (CBC). The peripheral blood smear shows Pappenheimer bodies, which are consistent with sideroblastic anemia. Isoniazid-induced B_6 deficiency is the most likely cause of sideroblastic anemia in this patient, therefore, the best treatment is vitamin B_6 supplementation. Deferoxamine (A) and phlebotomy (B) would be effective for iron overload; this would be uncommon in a patient with acquired sideroblastic anemia and is seen mostly in hereditary (genetic) causes. Vitamin B_{12} (D) deficiency causes megaloblastic anemia, which is a macrocytic anemia. Folate deficiency (E) causes megaloblastic anemia.

3. A 2-month-old male was brought to the physician for failure to thrive that began shortly after birth. Subsequent investigations in the neonate confirm sideroblastic anemia. Which of the following is the most likely cause of his condition?

A. ○ Dysplastic cells in the bone marrow
B. ○ Decreased ferrochelatase activity
C. ● Decreased
D. ○ Maternal iron deficiency during pregnancy
E. ○ Vitamin B_6 deficiency

Hide Explanation

The correct answer is decreased ALAS activity (C). This is a case of congenital sideroblastic anemia given the patient’s age and onset of symptoms shortly after birth. The disease is an X-linked gene mutation causing deficient δ-aminolevulinic acid synthase (ALAS) activity, needed for heme synthesis. Dysplastic cells in the bone marrow (A) are seen in myelodysplastic syndrome, which is more likely seen in adults and is not a congenital cause of sideroblastic anemia. Decreased ferrochelatase activity (B) is seen in lead toxicity. Maternal iron deficiency during pregnancy (D) is not a cause of sideroblastic anemia. Vitamin B_6 deficiency (E) is an acquired cause of sideroblastic anemia that results from severe malnutrition, alcohol use, or isoniazid use.

4. In a patient with sideroblastic anemia, which of the following findings is least suggestive of the development of secondary hemochromatosis?

A. ○ Bilateral peripheral pitting edema
B. ● Conjunctival pallor
C. ○ Elevated liver enzymes
D. ○ Excess th
E. ○ Hyperpigmented skin

Hide Explanation

The correct answer is conjunctival pallor (B). In sideroblastic anemia, conjunctival pallor is due to low hemoglobin not iron overload. Secondary hemochromatosis, which means acquired iron overload, is a complication of sideroblastic anemia due to ineffective erythropoiesis in the bone marrow. Excess iron typically deposits in the heart, skin, pancreas, and liver. Cardiac involvement leads to heart failure, which presents with bilateral peripheral pitting edema (A). Excess hepatic iron leads to inflammation, elevated liver enzymes (C), abdominal pain, and (chronically) liver cirrhosis. Longstanding pancreatic injury results in symptoms of diabetes mellitus such as excess thirst (D) and urination. Iron deposits in the skin stimulate melanocytes leading to hyperpigmented skin (E).

How would you rate this Brick?

References

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References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.
- Hoffman R, Benz EJ, Silberstein L, et al. *Hematology: Basic Principles and Practice*. 8th ed. Elsevier; 2023.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☐ **Differentiate the clinical presentation of the mentioned anemia.**
02 ☐ **Describe the pathophysiology of the mentioned anemia.**
03 ☐ **Interpret clinical findings to properly diagnose the mentioned anemia.**
04 ☐ **Outline the management of the mentioned anemia.**

Objective Confidence: 0/4